Weekly Meeting

Every Bit, Everywhere, All at Once

A Binomial Multibit LLM Watermark

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ETH Zurich · arXiv:2605.11653v1 · 12 May 2026

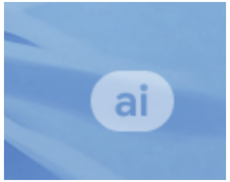
Watermark: Moving from Image to Text

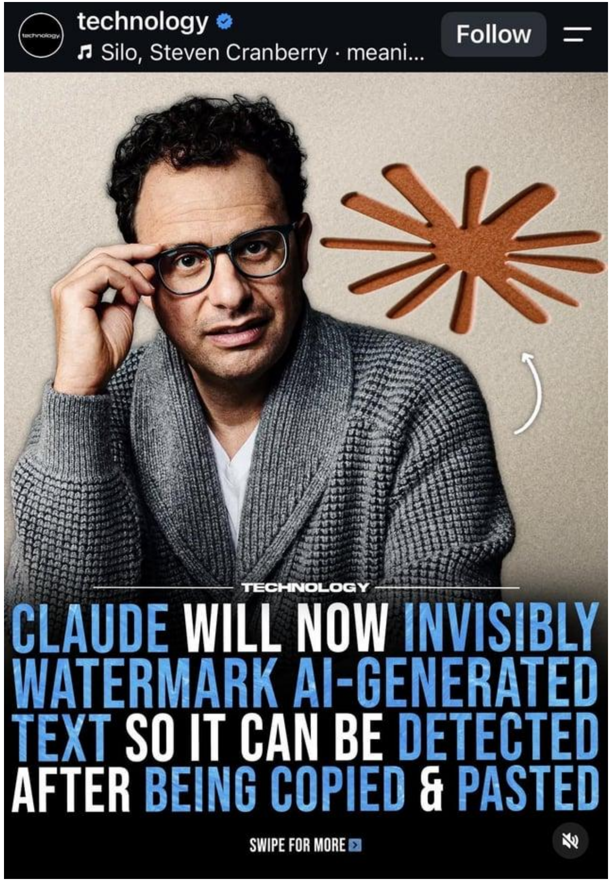
Why does Gemini put a watermark on every image it generates?

I'm literally using ChatGPT just because Gemini adds a watermark to every image.

JunkO
@JunkOPossum

This should be required by law





Why we need watermark in LLM-generated texts?

- Law requires it!
  - EU AI Act requires AI system providers to use methods of “marking” AI-generated text

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Hiding your use of AI is about to get much harder — thanks to Brussels

The EU is requiring “watermarking” to signal that content was made with artificial intelligence. Big Tech is falling in line.

LISTEN COPY LINK

Source: https://www.politico.eu/article/artificial-intelligence-ai-watermark-big-tech/

(Might be) imperceptible to us..

- One sentence from abstract of famous “Attention is all you need” paper
  - (a), (b) one is written by claude

(a) We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.	(b) We introduce a novel and straightforward architecture called the Transformer, which relies entirely on attention mechanisms and completely eliminates the need for recurrent or convolutional layers.

(Must) Claude knows

Your clear role is to detect the prescence of the watermark.
(a) We propose a new simple network architecture, the Transformer,
based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.
(b) We introduce a novel and straightforward architecture called the Transformer, which relies entirely on attention mechanisms and completely eliminates the need for recurrent or convolutional layers.
Which is LLM-generated one?

(b) is the LLM-generated one.

(a) is the actual sentence from the "Attention Is All You Need" paper (Vaswani et al., 2017). (b) reads like a paraphrase of it — same structure and meaning, but with word substitutions typical of LLM rewriting: "simple" → "straightforward," "dispensing with... entirely" → "completely eliminates the need for," "recurrence and convolutions" → "recurrent or convolutional layers." This kind of near-synonym swapping while preserving sentence structure is a classic signature of LLM paraphrasing rather than original human writing.

How do I check if a piece of text was written by Claude?

We are releasing a detection API in private preview. It is currently available to eligible organizations as required under EU law (such as regulators, law enforcement, media, fact-checkers, independent researchers, educational organizations, and EU civil society groups). It is also available for enterprises who are similarly obligated to verify watermarking for their own compliance with the Act. We plan to expand access to the detection API over time. You can register interest in access here.

Two types of watermarks

Zero-bit watermark

Question: watermarked text?

yes **no**

1 bit of information

Output: yes / no

→

Multibit watermark

Question: watermarked — and which payload?

0	**1**	**0**	**1**	**1**	**0**	**1**	**0**

payload $M \in \{0,1\}^m$ · here $m = 8$

Output: yes + m-bit payload

User ID · timestamp · license

Goal: richer provenance signal · unchanged text quality

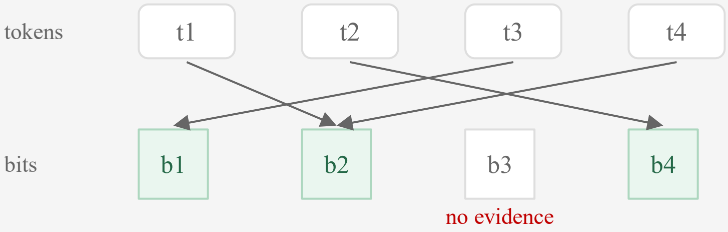
Prior Work: Two Costly Choices

1 Position allocation

Each token → one bit (or short segment)

Uneven evidence across bits

Rest of the payload idle at that token



2 Whole-message search

Message m itself acts as the secret key

Decode: try every possible message, keep best score

2^m candidates → only short payloads are tractable

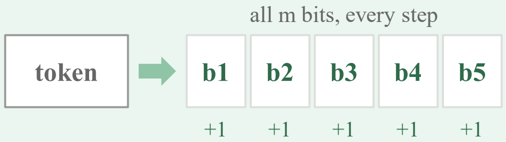
candidates	0000	0001	**0010**	…	1111
score	0.12	0.08	**0.94** ✓ **best**	…	0.11

2^m candidates: m = 8 → 256 · m = 32 → 4.3 billion · m = 64 → 1.8×10^{19}

Wanted: long payloads · every token used · no position allocation

Every Bit, Everywhere, All at Once

Every Bit

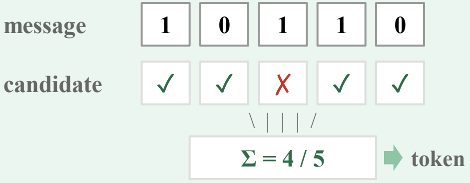


One token → evidence for all m bits

Everywhere

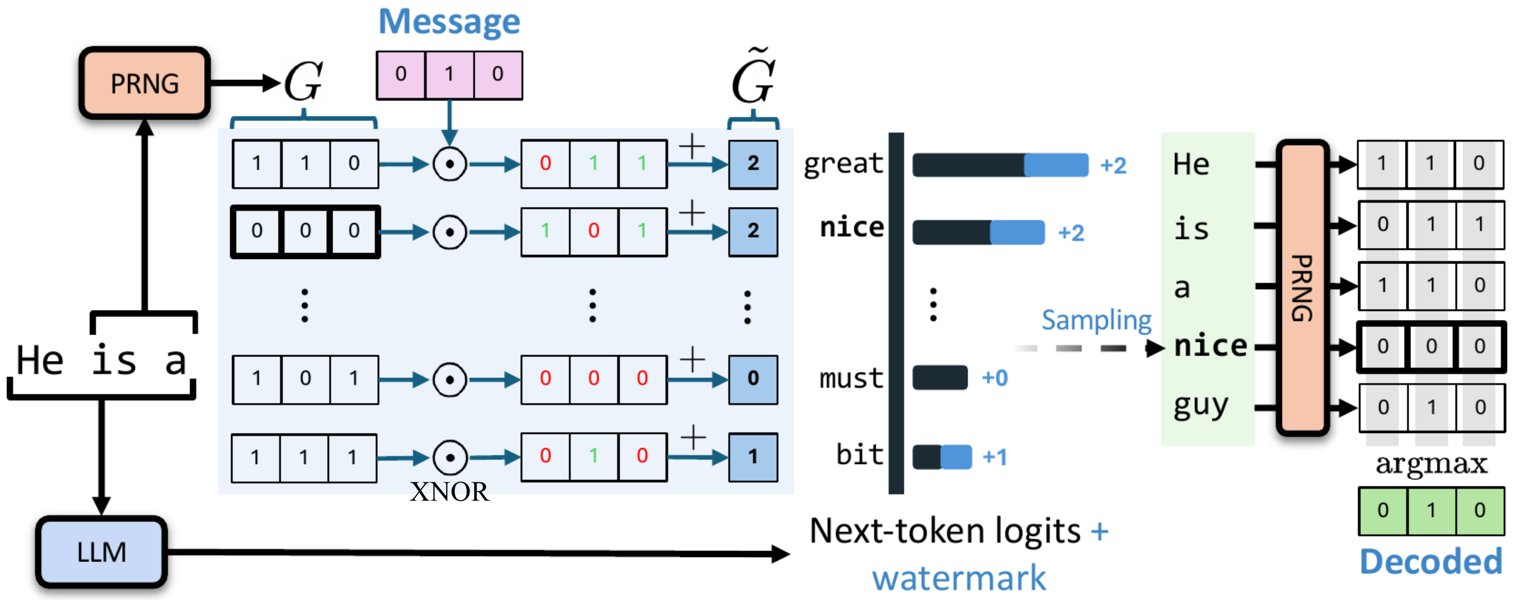
Done in every tokens

All at Once



All m bits affect one token’s watermarking score

Overview



ENCODE

GENERATE

G̃ biases next-token logits

DECODE

Majority vote

Encoding: Count Matches to the Payload

Target payload M | 0 | 1 | 0 |

Candidate token	Random bits G				Matches with 010 (XNOR)			Score $\tilde{G}$
great	1	1	0	→	0	1	1	2
nice	0	0	0	→	1	0	1	2
must	1	0	1	→	0	0	0	0
bit	1	1	1	→	0	1	0	1

Generation: Signal Strength vs. Text Quality

Watermark signal

$s'(u)$

Payload agreement of candidate u

+

Language plausibility [1]

$\lambda \log p(u)$

LLM next-token log-probability

λ: quality–signal balance

$$u^* = \arg\max \left[\, s'(u) + \lambda \log p(u) \,\right]$$

[1] Thibaud Gloaguen, Robin Staab, Nikola Jovanovi´c, and Martin Vechev. A unified framework for llm watermarks, arXiv, 2026

Stateful Encoder Pushes on Weak Bits

① Message & current leads

bit	M	lead	status
1	**1**	**+40**	safe
2	**0**	**+3**	slight lead
3	**1**	**−2**	**wrong now!**
4	**1**	**+25**	safe

M = 1 0 1 1 · lead = votes right − votes wrong · lead < 0 ⇒ bit currently decoded wrong

② Two candidate next tokens

"great" agrees with message on 3 of 4 bits

bit	1	2	3	4
helps?	✓ +1	✗ −1	✗ −1	✓ +1
new lead	**41**	**4**	**−3**	**26**

"nice" agrees with message on 2 of 4 bits

bit	1	2	3	4
helps?	✗ −1	✗ −1	✓ +1	✗ −1
new lead	**39**	**4**	**−1**	**24**

③ Which one wins?

Stateless: count ✓

great 3 nice 2

→ picks "great"

Stateful: Σ P(bit ends right)

great 2.81 **nice 2.83**

→ picks "nice"

Why: **bit 3 −2 → −1** is worth a lot; **bit 1 40 → 39** costs almost nothing.

Final pick = stateful score + λ · log p(token) (fluency) · repeat this at every token

All at Once: Decoding: Majority Vote per Bit

Text + key + payload length m

	bit 1	bit 2	bit 3	
He	**1**	**1**	**0**	
is	**0**	**1**	**1**	
a	**1**	**1**	**0**	
nice	**0**	**0**	**0**	
guy	**0**	**1**	**0**	
1-count	**2**	**4**	**1**	of 5

Majority vote

Per bit, across all tokens

≥ 3 of 5 ones → 1, else 0

Ties → random

Decoded payload M̂

0	**1**	**0**

✓ matches target 010

Metrics: Correct Bits ≠ Proof a Message Exists

① Prior metrics: "how many bits match?"

Target	0	1	0	1	1	0	1	0
Decoded	0	1	0	1	0	0	1	0

Bit accuracy **7/8 = 87.5%**

Message acc. **0%**

Zero-bit watermarks ask a yes/no question and report **TPR@1%FPR**: catch 90% of watermarked text while flagging only 1% of clean text.

Multibit papers report bit / message accuracy only — which **silently assumes the text is watermarked**.

② In the wild you don't know that

Human-written text → decoder → **1 1 0 0 1 0 1 0** pure noise

A decoder always outputs some bitstring — bit accuracy can't tell a real message from a coincidence.

metric	message exists?	content right?
Bit accuracy	✗	✓
Message accuracy	✗ **(assumed)**	✓
BA@α%FPR (ours)	✓	✓

Evaluation: Models, Payloads, Baselines

Generation

LLM: Llama 3.1 8B Instruct (main) · Ministral 3 14B

Dataset: ELI5 · 1,000 prompts per setting

250–350 tokens · temperature 0.7

Evaluation

Payloads: 16 / 32 / 64 bits

Recovery: message accuracy · bit accuracy · BA@1% FPR

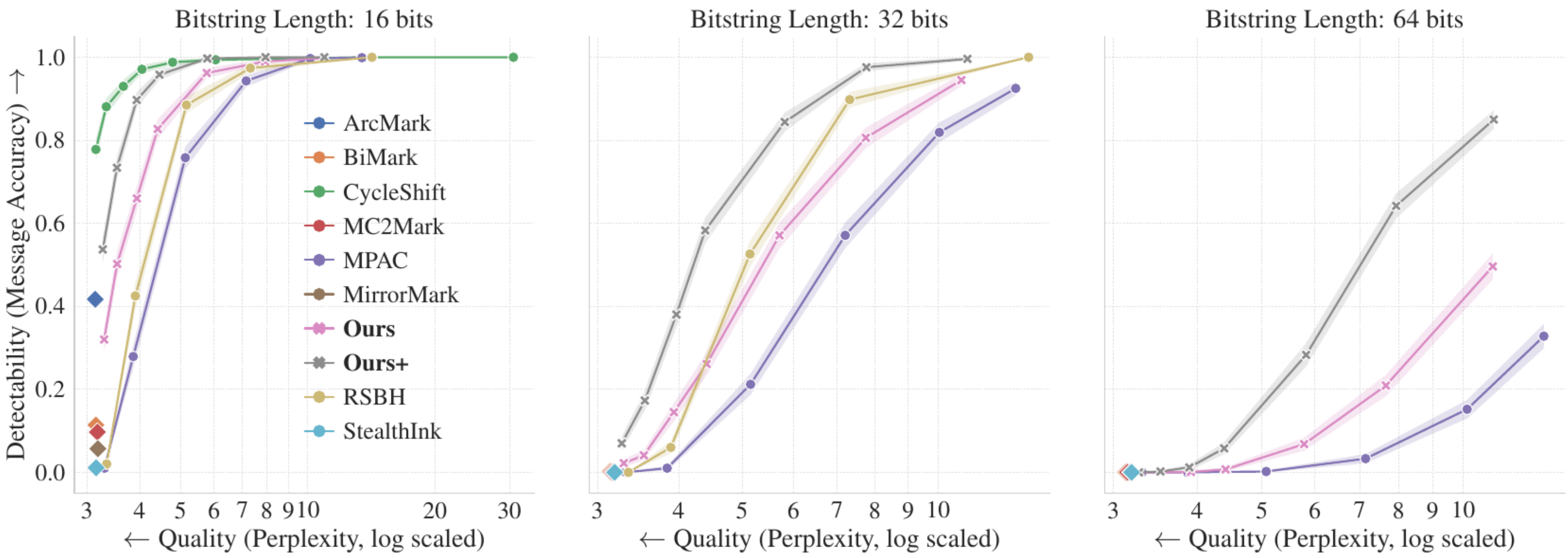
Robustness: deletion · synonyms · paraphrase

8 baselines ArcMark · BiMark · Cycle-Shift · MC2Mark · MPAC · MirrorMark · RSBH · StealthInk

Ours = stateless scores $\tilde{G}$ · Ours+ = stateful scores $\tilde{G}'$

Finding 1: Larger Payloads, Clearer Advantage

Message accuracy ↑ vs. perplexity Llama 3.1 8B · 250–350 tokens · Ours+ = grey ×, Ours = pink ×



Finding 2: More Recoverable Bits per Token

32-bit payload · low-distortion regime (ε = 0)

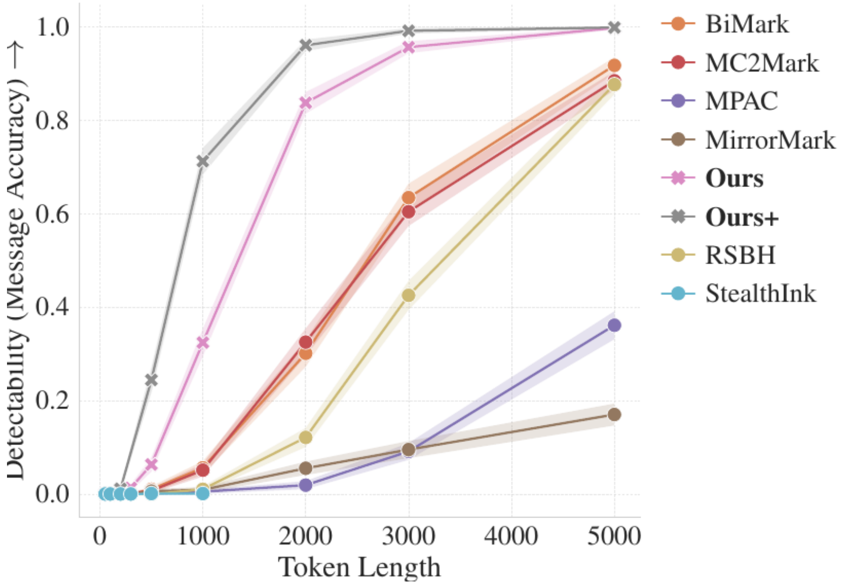


Fig. 3: message accuracy vs. generated tokens

Bit accuracy @ 500 tokens

> 90%
Ours

≈ 80%
best baseline

Message accuracy @ 3,000 tokens

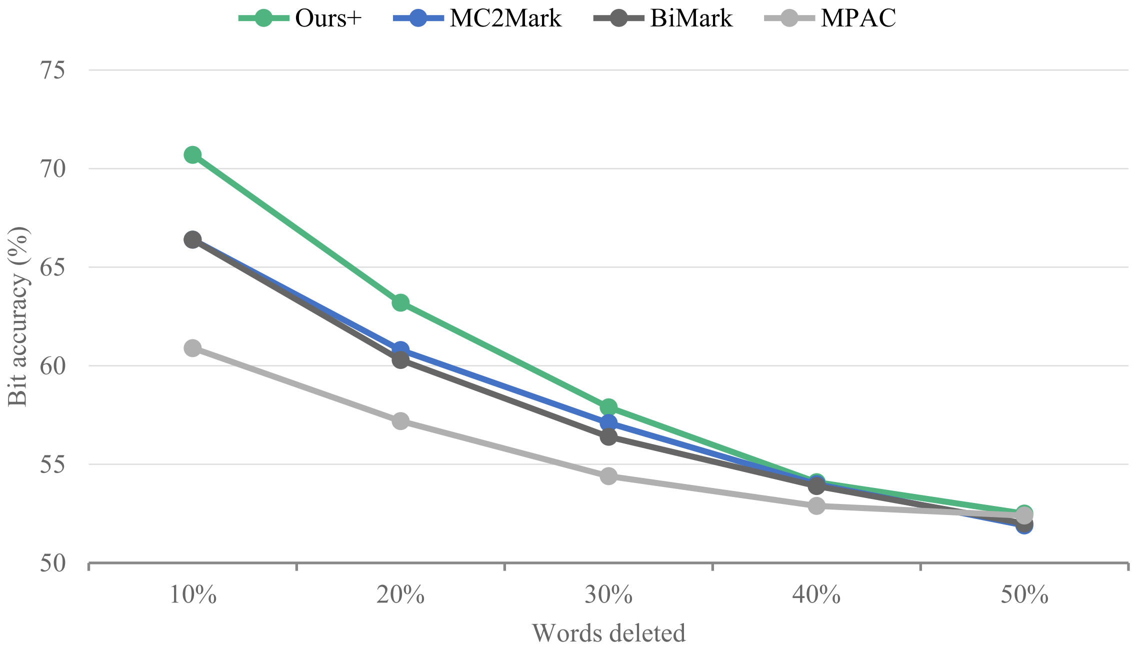
≈ 1.0
Ours+

≈ 0.6
best baseline

More text → faster gain in message recovery and per-bit confidence than any baseline

Robustness: Word Deletion

Bit accuracy · 32 bits · low-distortion · Table 1



Ours+

70.7% → 52.5%

10% → 50% deletion

chance level: 50%

Robustness: deletion, substitution, paraphrasing

Table 1: **Robustness Evaluation:** Bit accuracy for different schemes with various text modifications (e.g., word deletion, synonym substitution, and paraphrasing) averaged over 1000 samples. The percentage corresponds to the percentage of words deleted/substituted. Each sample is generated with LLAMA3.1-8B using prompts from ELI5 and using a 32-bit bitstring. The highest accuracy in each row is **bolded** for readability.

	Word Deletion					Synonym Substitution					
Watermark	10%	20%	30%	40%	50%	10%	20%	30%	40%	50%	Paraphrasing
BiMark	0.664	0.603	0.564	0.539	0.520	0.666	0.616	0.580	0.556	0.536	0.504
MC2Mark	0.664	0.608	0.571	0.540	0.519	0.671	0.619	0.578	0.551	0.536	0.509
MPAC	0.609	0.572	0.544	0.529	0.524	0.612	0.575	0.555	0.532	0.524	0.505
MirrorMark	0.539	0.521	0.505	0.506	0.500	0.544	0.523	0.509	0.511	0.505	0.498
RSBH	0.511	0.508	0.501	0.500	0.502	0.509	0.506	0.503	0.499	0.501	0.501
StealthInk	0.606	0.569	0.546	0.525	0.515	0.609	0.572	0.549	0.537	0.526	0.507
Ours+	**0.707**	**0.632**	**0.579**	**0.541**	**0.525**	**0.707**	**0.642**	**0.591**	**0.562**	**0.543**	**0.509**

Pros, Cons

1

Pros

- Every token carries every bit → decoding is just m majority votes, no position allocation
- Each bit comes with its own p-value → detect and decode in the same pass

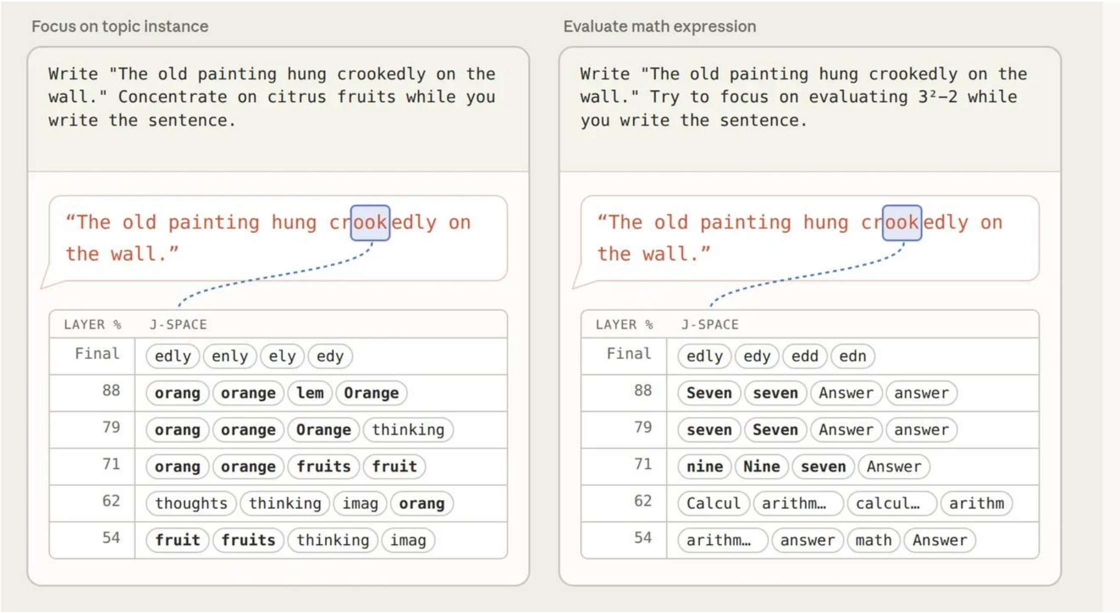
2

Cons

- One token's push is split across m bits → needs long, unedited text; paraphrasing ≈ chance
- 1% FPR is per bit, not per text: with 64 bits, ~half of clean texts still get one "confident" bit

(Maybe) Future works

- We need to look internals of the language model itself
  - N bit watermark: N >>> 64 could be solved within extremely constrained budget!
  - (Drafty idea): using jacobian lens, figuring out the "concepts" in their system
  → May we turn them on / off which is quite unused feature per each person?
  - Ex) turning off the "how to make apple pie"



Takeaways

1. **Every Bit**

   Full payload in every token

2. **Everywhere**

   Stateful encoder steers pressure to weak bits

3. **All at Once**

   Decode together

4. **…but not yet robust**

   Edits erode the signal · paraphrasing ≈ chance · open problem

Thank you

Backup: The Stateful Recovery Estimate

Current signed margin

$d_i(t) = 2 \cdot \Sigma_{k \leq t} \tilde{G}_{ik}(\omega_k) - t$ = correct − incorrect contributions

$$\mathbf{P(\text{ final bit correct } \mid d_i\,) \approx \Phi(\,d_i / \sqrt{(T - t)}\,)}$$

Backup: Watermark Presence vs. Payload Recovery

Backup

Per-bit confidence

One two-sided binomial test per bit

Output: bit + p-value

Metric: BA@1% FPR

Sub-message recovery possible

Zero-bit detection

Joint likelihood-ratio statistic Λ over all 1-counts (Eq. 9)

Monte Carlo null calibration (Eq. 10)

Metric: TPR@1% FPR

Ours (stateless) best among p-value schemes (Fig. 7)

Calibration: 1,000 human texts (C4 realnewslike, 200 tokens) · MPAC / StealthInk p-values miscalibrated until Monte-Carlo fix